

Power Connections

From	To	Wire Color Suggestion
TP4056 OUT+	ESP32 VIN	Red
TP4056 OUT-	ESP32 GND (either one)	Black
TP4056 OUT+	LCD Pin 2 (VDD)	Red
TP4056 OUT-	LCD Pin 1 (VSS)	Black

Voltage Divider (Battery Sense)



From	To
TP4056 OUT+	R1 leg 1
R1 leg 2	R2 leg 1 + ESP32 D34
R2 leg 2	GND rail

LCD Connections

Your board has D13, D12, D14, D27, D26, D25 all on the left side — perfect, all LCD data pins are grouped together.

LCD Pin	Label	Connects To
Pin 1	VSS	GND rail
Pin 2	VDD	TP4056 OUT+
Pin 3	V0	40kΩ resistor → GND (<i>contrast fix</i>)
Pin 4	RS	ESP32 D13
Pin 5	RW	GND rail
Pin 6	E	ESP32 D12
Pin 7	D0	Not connected
Pin 8	D1	Not connected
Pin 9	D2	Not connected
Pin 10	D3	Not connected
Pin 11	D4	ESP32 D14
Pin 12	D5	ESP32 D27
Pin 13	D6	ESP32 D26
Pin 14	D7	ESP32 D25
Pin 15	A (Anode)	470Ω resistor → ESP32 D2
Pin 16	K (Cathode)	GND rail

All GND Points — Wire Them All Together

Your board has two GND pins — use both if needed or daisy chain:

- ESP32 GND
- LCD Pin 1
- LCD Pin 5 (RW)
- LCD Pin 16 (K)
- ~~10k contrast resistor bottom leg~~
- Voltage divider R2 bottom leg
- All three button second legs

COUNTDOWN TIMER - V1

